

Software Requirements Specification

for

Microbanking and Interest Management System

Version 1.0 approved

Prepared by Group 31

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the Microbanking and Interest Management System (MIMS), developed for B-Trust, a microfinance bank in Sri Lanka. It covers the backend database design and the lightweight QA web interface used to manage customer registration, savings accounts, deposits/withdrawals, fixed deposits, and automated monthly interest calculation. Future channels such as mobile banking, SMS/USSD, or payment gateway integrations are out of scope

1.2 Document Conventions

- Database object names follow standard naming prefixes: sp_ for stored procedures, fn_ for stored functions, and trg_ for triggers.
- All monetary values are expressed in Sri Lankan Rupees (LKR) unless otherwise stated.
- Priority levels (High, Medium, Low) assigned to each System Feature are inherited by all functional requirements listed beneath it, unless a requirement explicitly states its own priority.

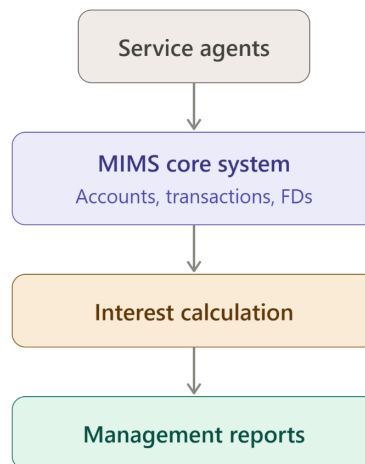
1.3 Intended Audience and Reading Suggestions

- **Development Team / Database Engineers** – to design and implement the schema, stored procedures, functions, and triggers described in Sections 3 and 4.
- **QA Testers** – to understand expected system behavior, validation rules, and error conditions for test case design, primarily using Sections 4 and 5.
- **Branch Managers and Banking Agents (future end users)** – to understand the operational workflows the system supports.
- Readers are advised to begin with Section 2 (Overall Description) for a high-level understanding of the system, then proceed to Section 3 (External Interface Requirements) for interface and technical stack details, followed by Section 4 (System Features) for detailed functional behavior, and Section 5 (Other Nonfunctional Requirements) for performance, security, and quality constraints.

1.4 Product Scope

- MIMS is a backend-driven microbanking database system built to digitize B-Trust's core operations.
- Core operations covered: customer and savings account registration, deposit/withdrawal transaction processing, fixed deposit management, and automated monthly interest distribution.
- Supports five savings plans: Children, Teen, Adult, Senior, Joint.
- Supports three fixed deposit tenures: 6 months, 1 year, 3 years.

- Supports B-Trust's goal of improving financial inclusion for rural communities by enabling regional service agents to manage customer accounts efficiently and accurately.
- Enforces strict data integrity, overdraft prevention, and ACID-compliant transaction processing.
- Includes a lightweight web-based QA console for testers/administrators to validate operations and generate the five mandatory management reports, without needing a full production-grade front end.



1.5 References

- I. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Computer Society.
URL: [IEEE Xplore Standards](#)
- II. ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering* (Superseded IEEE Std 830).
URL: [ISO/IEC/IEEE 29148 Standard](#)
- III. Personal Data Protection Act, No. 9 of 2022, Parliament of the Democratic Socialist Republic of Sri Lanka.
URL: [Sri Lanka Parliament Acts](#)
- IV. *MySQL 8.0 Reference Manual*, Oracle Corporation (Stored Procedures, Triggers, and InnoDB Transaction Model).
URL: [MySQL 8.0 Reference Documentation](#)
- V. Node.js mysql2 Database Client Library, GitHub Community.
URL: [mysql2 Driver Documentation](#)
- VI. Central Bank of Sri Lanka (CBSL), *Microfinance Act, No. 6 of 2016 & Operational Guidelines for Microfinance Institutions*.
URL: [Central Bank of Sri Lanka Regulations](#)
- VII. Lucidchart Diagramming & Flowchart Software, Lucid Software Inc. (used for context, DFD, and class diagrams).
URL: [Lucidchart Official Site](#)
- VIII. Mermaid Live Editor, Mermaid.js Community (used for state-transition diagrams and ER database modeling).
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2. Overall Description

2.1 Product Perspective

The Microbanking and Interest Management System (MIMS) is a new, self-contained backend database system, together with a lightweight validation UI, commissioned by B-Trust, a small private microfinance bank operating across several districts in Sri Lanka. MIMS is not a follow-on to an existing automated system, it replaces what is currently a manual, paper-based process for registering customers, tracking Savings Accounts and Fixed Deposits, and calculating interest. MIMS is not a component of any larger corporate system and has no dependency on an external database or third-party service.

Because most B-Trust customers live in rural districts with limited direct access to computing devices, the **Customer** does not connect to the system directly, instead, a **Banking Agent** at a regional Branch performs registration, deposits, withdrawals, and Fixed Deposit transactions on the Customer's behalf, and the Customer may view a read-only balance notice through the Agent. The **QA Tester** connects to a lightweight web interface that talks directly to the backend database in order to validate core operations, stored procedures, functions, and triggers. The **Interest Calculation Engine** is not a human actor but an automated, scheduled process inside the system that calculates and posts interest on a 30-day cycle without requiring direct human interaction. The diagram below shows the major components of the system and how these actors interconnect with it.

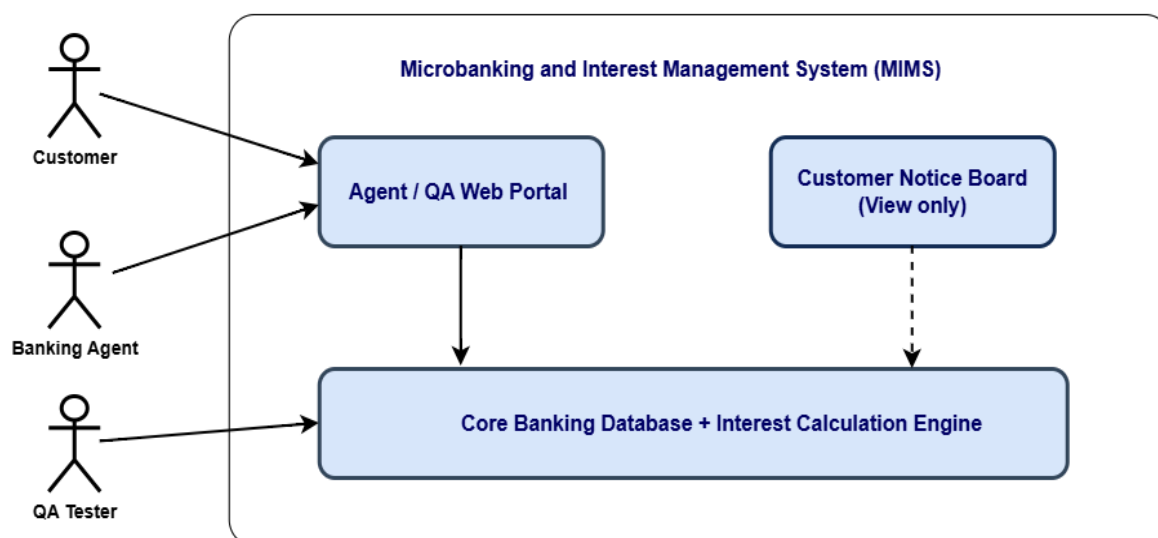


Figure 2.1 - System Context Diagram

B-Trust operates a small number of service Branches, each of which manages a team of Banking Agents. Every Customer registers at a Branch and is assigned to a specific Agent, who remains the Customer's primary point of contact for all subsequent transactions.

2.2 Product Functions

The major functions MIMS must perform, or must let an Agent perform on a Customer's behalf, are summarised below. Detailed functional requirements for each are provided in Section 3.

- **Customer and Agent Registration** — register Customers at a Branch and assign each Customer to a responsible Agent.
- **Savings Account Management** — open and maintain Savings Accounts for individual, minor, senior, or joint Customers under one of five Account Plans.
- **Deposit and Withdrawal Processing** — record deposits and withdrawals during business hours, enforcing minimum-balance and no-overdraft rules.
- **Fixed Deposit Management** — open a Fixed Deposit linked to an eligible Savings Account under one of three Terms, limited to one active Fixed Deposit per Savings Account.
- **Automated Interest Calculation and Crediting** — calculate and post Fixed Deposit interest to the linked Savings Account on a recurring 30-day cycle, without manual intervention.
- **Transaction Logging** — log every deposit, withdrawal, and interest credit with a timestamp, transaction type, and a system-generated reference number.
- **Management Reporting** — produce agent-wise transaction totals, account-wise summaries, active-FD listings, monthly interest distribution, and customer activity reports.
- **Database Validation Interface** — provide a lightweight UI through which a QA Tester can directly exercise and verify the underlying stored procedures, functions, and triggers.

The Savings and Fixed Deposit products that these functions must support are summarised in the tables below, as they define the eligibility and validation rules enforced throughout the requirements in Section 3.

Savings Plan	Interest Rate	Minimum Balance	Eligibility
Children	12%	None	Customer under 13 years
Teen	11%	LKR 500	Customer 13-17 years
Adult	10%	LKR 1,000	Customer 18-59 years
Senior	13%	LKR 1,000	Customer 60 years and above
Joint	7%	LKR 5,000	Two or more account holders

Fixed Deposit	Term Interest Rate	Interest Posting
6 months	13%	Monthly, to linked Savings Account

1 year	14%	Monthly, to linked Savings Account
3 years	15%	Monthly, to linked Savings Account

A Customer may hold only one active Fixed Deposit per Savings Account, and every Fixed Deposit must be linked to an active Savings Account belonging to the same Customer. Each monthly interest credit, and every deposit or withdrawal, is recorded as an individually logged transaction.

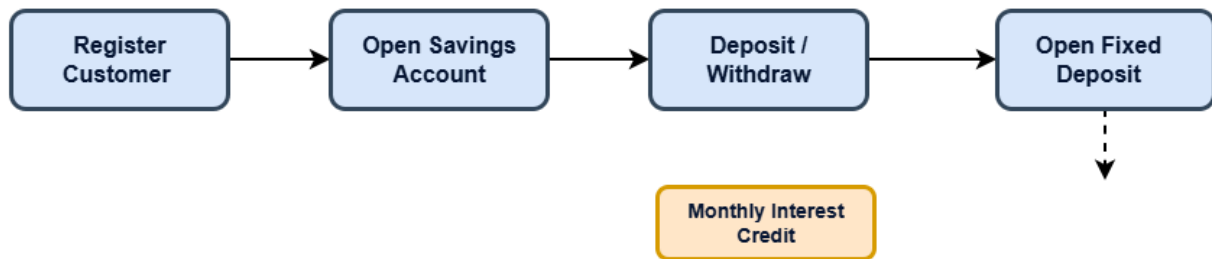


Figure 2.2 - Savings and Fixed Deposit Lifecycle

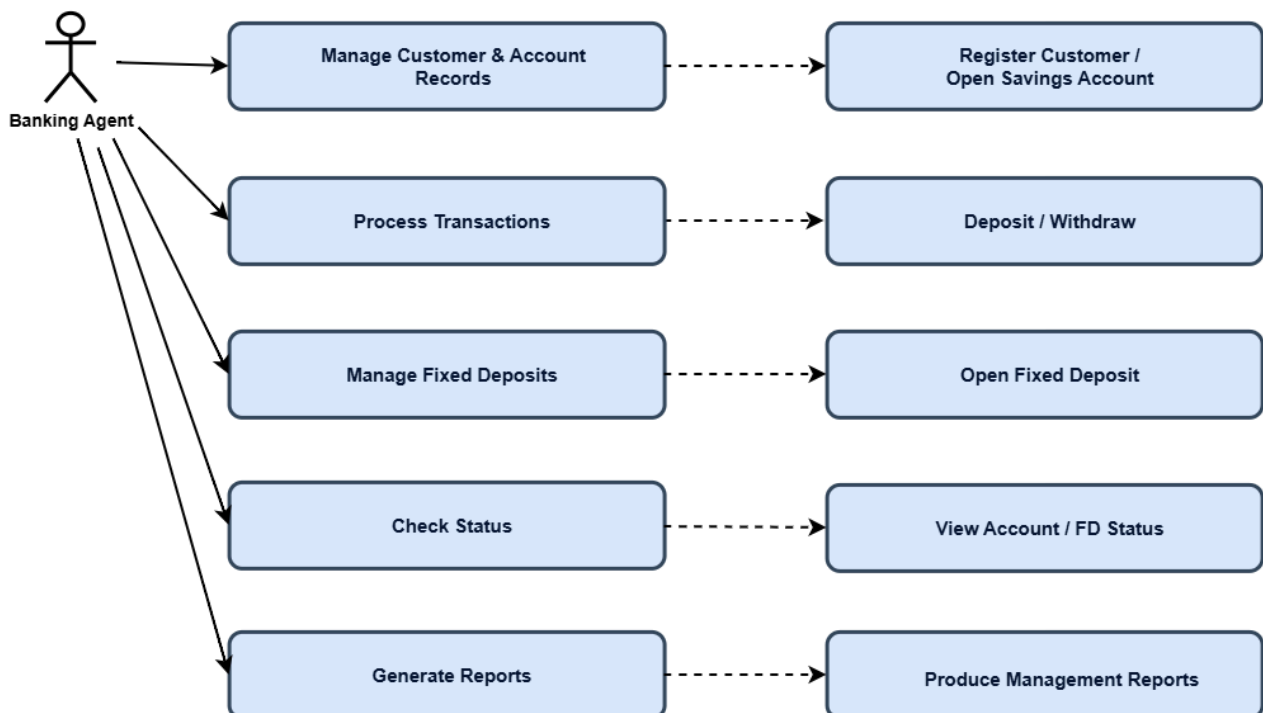


Figure 2.3 - Grouping of Banking Agent Functions

A Customer registers at a Branch and is assigned to an Agent, who opens one or more Savings Accounts on the Customer's behalf. Deposits and withdrawals are processed against those accounts throughout the account's life. Once a Savings Account is active, the

Customer may ask the Agent to open a linked Fixed Deposit, which then generates its own monthly interest-credit transactions back into the Savings Account until maturity.

2.3 User Classes and Characteristics

The user classes anticipated for MIMS, and their pertinent characteristics, are summarised below. The Banking Agent is the most important user class to satisfy, since almost all data-entry and transaction-processing activity is performed by an Agent on the Customer's behalf.

User Class	Frequency of Use	Frequency of Use	Privilege Level
Customer	Occasional (via Agent)	None assumed not expected to interact with the system directly.	None — read-only balance notice communicated through the Agent.
Banking Agent	Daily, primary user	Comfortable with a standard web browser, forms, and grids; no database or programming background required.	Register Customers open/manage Savings Accounts and Fixed Deposits; process deposits and withdrawals generate reports for assigned Customers.
Branch Manager	Daily to weekly	Same as Banking Agent.	Same as Banking Agent, plus reports summarising activity across an entire Branch.
QA Tester	During test cycles only	Technically proficient; familiar with relational database concepts (keys, transactions, constraints); no microfinance domain expertise required.	Access limited to the test interface; may exercise stored procedures, functions, and triggers but must not bypass data-integrity constraints.
Interest Calculation Engine (system actor)	Automated, every 30-day cycle	Not applicable — automated backend process.	System-level; no human privilege boundary.

2.4 Operating Environment

The Core Banking Database and Interest Calculation Engine are hosted on a relational database management system (RDBMS) capable of enforcing primary/foreign key constraints, stored procedures, functions, and triggers (for example MySQL), running on server hardware provided by, or on behalf of, B-Trust. The Agent Portal and the QA Tester's lightweight validation UI are web-based and are accessed through a standard web browser at each Branch or test location, requiring only a stable internet or intranet connection to the database server; no special client hardware or operating system is required beyond a modern browser. The system must coexist with, but has no functional dependency on, any other software already in use at B-Trust's Branches.

2.5 Design and Implementation Constraints

Constraint	Description
Data Integrity (ACID)	Every deposit, withdrawal, Fixed Deposit opening, and interest credit must be processed as an atomic, consistent, isolated, and durable transaction, implemented through stored procedures and triggers rather than relying solely on application-level checks.
No Overdrafts	Withdrawals that would reduce a Savings Account below its Plan's minimum balance, or below zero, must be rejected by the database layer itself.
Referential Integrity	Primary and foreign keys must enforce the relationships between Branches, Agents, Customers, Savings Accounts, Fixed Deposits, and Transactions, including the rule that a Savings Account may be linked to at most one active Fixed Deposit.
Performance / Indexing	Indexes must be created on frequently queried columns, including Account Number, Customer ID, Agent ID, and Transaction Date, to keep report generation and balance lookups responsive as transaction volume grows
Security	A QA Tester's access is limited to the test interface and must not be able to bypass the constraints and triggers that protect production data.
Currency and Locale	All monetary values are recorded and processed in Sri Lankan Rupees (LKR) no multi-currency support is required
Sample Data Volume	The initial database must be populated with at least 5 Agents, 3 Branches, 15 Customers (including at least 2 Joint Accounts), 10 Fixed Deposits, and 100 transactions, to support QA validation.

2.6 User Documentation

The following documentation will be delivered together with MIMS a database design document describing the entity-relationship model, table definitions, and the stored procedures, functions, and triggers that enforce the requirements in Section 3 a short QA Tester's guide describing how to use the lightweight validation UI to exercise each database operation and an annotated sample data set (Agents, Branches, Customers, Fixed Deposits, and Transactions) that can be used both for testing and as a worked reference for expected system behaviour. No end-user training manual is planned for the Customer, since Customers do not interact with the system directly brief in-app guidance is provided to Agents within the Agent Portal itself.

2.7 Assumptions and Dependencies

- Business hours, during which deposits and withdrawals may be processed, are assumed to be the standard working hours of each Branch (approximately 9:00 AM to 5:00 PM, Monday to Friday). The exact schedule is configurable but not otherwise specified in the requirements.
- 30 days is assumed for every interest-calculation cycle, regardless of calendar month length, as stated in the project brief.
- No withdrawal limit beyond the Plan's minimum balance has been specified. MIMS assumes no fixed per-transaction or per-day withdrawal ceiling other than the minimum-balance and no-overdraft rules.
- MIMS is assumed to be a standalone system with no dependency on an external membership or credit-bureau database, unlike systems that verify membership against a separate organisation.
- The project assumes the selected RDBMS supports stored procedures, functions, and triggers sufficient to enforce ACID properties at the database layer; if a simpler engine without this support were substituted, several requirements in Section 3 would need to be re-implemented at the application layer instead.
- All account holders are assumed to be adult-supervised where the Customer is a minor (Children or Teen plan); the system does not independently verify guardianship, which is assumed to be confirmed by the Agent at registration.
- The bank is assumed to operate only in Sri Lanka and only in LKR no foreign-currency or cross-border transaction support is assumed.

3. External Interface Requirements

3.1 User Interfaces

The microbanking and Interest Management System (MIMS) provides a responsive, single-page Web User Interface (SPA) tailored for operational management, QA testing, and technical proposal demonstrations.

3.1.1 GUI Standards and Product design System

- **Color Palette and Visual Hierarchy:**
 - Primary Accent Color: Warm Golden Yellow
 - Container Color: Warm Charcoal
 - Background and Surface: Light neutral slate
- **Window Standard:** Every major interface component is enclosed within a window frame card featuring a clean top header bar with explicit title text (e.g., MIMS QA Console - Dashboard, Transaction Execution Panel, Fixed Deposit Creation Screen, Management Reporting Dashboard).
- **Sidebar Navigation Standards:** Warm charcoal fixed sidebar featuring rounded pill button navigation items.
- **Screen Layout Constraints:** Responsive layout optimized for desktop display (1920X1080 standard). The sidebar remains fixed on the left while the primary workspace area dynamically scrolls vertically.

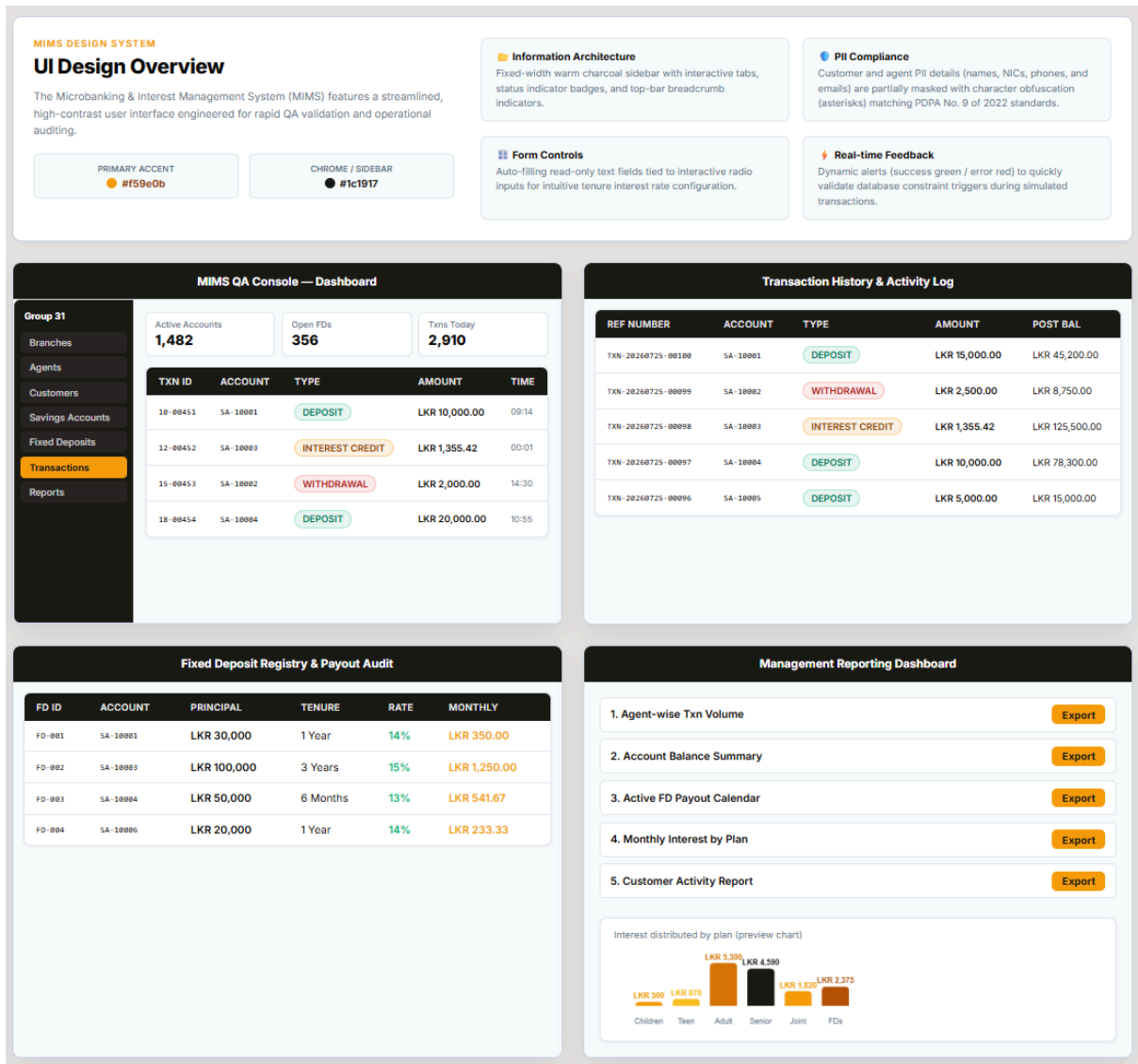
3.1.2 Standard Buttons, Controls and Feedback Display

- **Standard Action Controls:**
 - Refresh button: Triggers telemetry recount and UI re-sync.
 - Export: One-click yellow buttons on each management report row generating formatted datasets.
 - Run Monthly Interest Engine: Primary action control executing batch interest calculation across all active accounts.
- **Feedback & Validation Message Standards:**
 - Success Alerts: Light green container with green text confirming committed ACID transactions and procedural completion.
 - Error and Rollback Alerts: Light red container displaying explicit SQL error codes and failure justifications.
 - Informational Warnings: Light yellow dashed container displaying business constraint guidance.
- **Data Privacy & Redaction Standards (PII Compliance):**
 - All Customer and Banking Agent Personally Identifiable Information (PII) including legal names, NIC/Passport numbers, phone numbers, and email addresses are partially masked using character-level obfuscation (e.g., Ka***Pe*** for names, 077 ***5678 for phones, ka***@gmail.com for email, and 20041234***** for NICs) across all UI tables and report summaries in compliance with the *Sri Lanka Personal Data Protection Act (PDPA) No.9 of 2022*.

- Structural business data (Customer IDs, Account Numbers, Plan Types, Balances, Interest Rates, Timestamps, and Status Badges) remain fully legible.

3.1.3 Software Components Requiring User Interfaces

1. **Service Branch & Banking Agent Console:** Read-only audit tables for reviewing branch locations and assigned field officer allocations.
2. **Customer Master Registry Panel:** Audit table displaying registered individual and joint customers.
3. **Savings Account Registry & Plan Configurator:** Overview of 5 account plans with rate matrix, minimum balance indicators, and current savings account logs.
4. **Transaction History & Activity Log Console:** Comprehensive list of 100 sample transactions logged in the database, with real-time statistics cards.
5. **Fixed Deposit Lifecycle Screen:** Inspection interface with active fixed deposit records (10 FDs) and business constraint rule logs.
6. **Monthly Interest Engine Simulator:** Central control panel for batch interest calculation.
7. **Management Reporting Dashboard:** Tabbed view displaying the 5 mandatory reports with HTML5 Canvas interest distribution charts.
8. **Wireframe Collage Composite:**



3.2 Hardware Interfaces

3.2.1 Server Hardware Interfaces

- **CPU:** Standard 2-Core or 4-Core 64-bit Processor (Intel Core i3 / i5, AMD Ryzen 3 / 5).
- **System Memory:** Minimum 4 GB RAM (8 GB recommended)
- **Storage System:** Minimum 20 GB of available hard disk drive (HDD) or solid state drive (SSD) storage for database tables and application assets.
- **Network Interface Card (NIC):** Standard 100 Mbps / 1Gbps Ethernet controller supporting TCP/IP socket binding on Port 3306 and Port 80/443.

3.2.2 Client Hardware Interfaces

- **Device Types:** Standard Desktop PCs or Laptops with resolution 1920x1080.
- **Input Interfaces:** Standard Keyboard and Mouse/Touchpad pointer interactions for input form navigation and button triggering.
- **Display interaction:** Standard Integrated Graphics Processor (Intel UHD/Iris Graphics, AMD Radeon Graphics) supporting HTML5 canvas and standard CSS transition rendering in modern browsers.
- **Peripheral Interfaces:** Standard printing capability via virtual Pdf printer (Print to PDF) for transaction receipt validation and management report exports.

3.3 Software Interfaces

3.3.1 Database Management System Interface (MySQL 8.0)

- **Database Engine:** MySQL 8.0
- **Storage Engine:** InnoDB Engine enforcing ACID transaction compliance, row level locking, foreign key cascading, and repeatable-read transaction isolation.
- **Database Object and Invocation Interface:**
 - Stored Procedures: Executed via CALL interface: sp_ProcessDeposit, sp_ProcessWithdrawal, sp_RunMonthlyEngine
 - Stored Functions: fn_CalculateMaturityDate returning DATE scalar values.
 - Database Triggers: trg_CheckSingleFDConstraint running before insert on Fixed_Deposit to abort insertion if an active FD already exists for the saving account.
- **Driver Integration:** Native Node mysql2 (or Python mysql-connector-python/PHP PDO_MYSQL) connection pool maintaining persistent TCP socket pools.

3.3.2 Operating System and Runtime Interfaces

- **Server Operating Systems:** Windows Server 2019/2022, Windows 10/11 Pro, Linux
- **Application Server Runtime:** Node.js / Express API framework, Python (v3.10+) FastAPI, or PHP (v8.1+).
- **Web Browser Runtimes:** Google Chrome, Microsoft Edge, Mozilla Firefox.

3.3.3 Shared Data & Communication Interfaces

- **Global Data Sharing Mechanisms:**
 - InnoDB Shared Buffer Pool for high-speed in-memory table caching.
 - Database Sequence / Auto-increment counters for unique Transaction Reference generation (TXN-YYYYMMDD-XXXXX).
 - Application Server session state and HTTP response payload caching.

3.4 Communications Interfaces

3.4.1 Client-Server Communication Protocol

- **Application Protocol:** TLS 1.3 (HTTPS) for communication between client web browsers and the Backend API Server.
- **API Architecture:** RESTful HTTP API with standardized verb routing (GET /api/reports, POST /api/transactions, POST /api/fixed-deposits, POST /api/interest-engine).
- **Payload Data Formatting:** Structured JSON for input parameter payload delivery and database procedure response objects.

3.4.2 Database Network Protocol

- **Transport Layer:** TCP/IP Socket communication over standard Port 3306 between the Backend API Server and the MySQL 8.0 Database Instance.
- **Data Transfer & Synchronization:** Asynchronous non-blocking I/O socket event loops handling database client requests with deterministic connection pooling and automated reconnect handlers.

3.4.3 Security, Encryption & Access Control

- **Channel Security:** TLS 1.3 / SSL encryption for external network traffic.
- **Query Security:** Mandatory parameterized SQL query binding to eliminate SQL Injection risks.
- **Role-Based Access Control (RBAC):** Session token verification distinguishing System Administrators, Branch Managers, and Banking Agents.

4. System Features

4.1 Customer and Savings Account Registration

4.1.1 Description and Priority

This feature allows regional service agents to register new individual or joint customers and open Savings Accounts under specific plans (Children, Teen, Adult, Senior, Joint).

- **Priority:** High (Core operational requirement).

4.1.2 Stimulus/Response Sequences

1. **Stimulus:** The Banking Agent navigates to the "Customer Master Registry Panel" and initiates a new customer registration.
2. **Response:** The system presents a data entry form.
3. **Stimulus:** The Agent inputs the customer details and assigns them to a specific branch/agent.
4. **Response:** The system validates the input, generates a unique Customer ID, and updates the database.
5. **Stimulus:** The Agent selects a Savings Account plan from the "Savings Account Registry & Plan Configurator".
6. **Response:** The system verifies age eligibility based on the selected plan, enforces minimum balance rules, and creates the account.

4.1.3 Functional Requirements

- **REQ-1.1:** The system shall allow an agent to register a customer as either an individual or a joint entity.
- **REQ-1.2:** The system shall restrict the creation of specific Savings Account plans based on age criteria (e.g., Children, Teen, Adult, Senior).
- **REQ-1.3:** The system shall partially mask Personally Identifiable Information (PII) on the UI, displaying names and identification numbers with character-level obfuscation.

4.2 Transaction Processing (Deposits & Withdrawals)

4.2.1 Description and Priority

This feature handles the core financial interactions, allowing customers to deposit and withdraw funds via their assigned agents during business hours. All operations must be strictly logged.

- **Priority:** High (Critical financial operation).

4.2.2 Stimulus/Response Sequences

1. **Stimulus:** The Agent accesses the "Transaction Execution Panel" and enters an account number and withdrawal/deposit amount.
2. **Response:** The system queries the current balance. If it is a withdrawal, the system verifies that the remaining balance will not drop below the plan's minimum required balance.
3. **Stimulus:** The Agent confirms the transaction.
4. **Response:** The system commits the transaction to the database, generates a unique transaction reference (e.g., TXN-YYYYMMDD-XXXXX), and updates the "Transaction History & Activity Log Console".
5. **Response:** The system displays a success alert (light green container) or an error/rollback alert (light red container).

4.2.3 Functional Requirements

- **REQ-2.1:** The system shall strictly prevent overdrafts; withdrawals must be rejected if they violate the minimum balance constraint of the specific Savings Account plan.
- **REQ-2.2:** The system shall log every transaction with a precise timestamp, transaction type (Deposit/Withdrawal), amount, and unique reference number.
- **REQ-2.3:** The system shall execute all deposits and withdrawals using ACID-compliant database transactions to prevent partial updates.
- **REQ-2.4:** The system shall automatically trigger a transaction rollback if any step of a deposit or withdrawal process is interrupted by a system or network failure, ensuring the account balance and transaction logs revert to their exact state prior to the initiation of the request.

4.3 Fixed Deposit Management

4.3.1 Description and Priority

This feature enables customers with an active Savings Account to open a Fixed Deposit (FD) for fixed tenures (6 months, 1 year, 3 years) at predefined interest rates.

- **Priority:** Medium.

4.3.2 Stimulus/Response Sequences

1. **Stimulus:** The Agent accesses the "Fixed Deposit Creation Screen" and inputs a target Savings Account.
2. **Response:** The system verifies if the account already has an active FD.
3. **Stimulus:** If no active FD exists, the Agent selects the tenure and principal amount, then submits.
4. **Response:** The system creates the FD record and updates the "Fixed Deposit Lifecycle Screen".

4.3.3 Functional Requirements

- **REQ-3.1:** The system shall restrict customers to a maximum of one (1) active Fixed Deposit per Savings Account.
- **REQ-3.2:** The system shall calculate the maturity date and expected interest based on the selected tenure (6 months at 13%, 1 year at 14%, 3 years at 15%).

4.4 Automated Monthly Interest Calculation

4.4.1 Description and Priority

This batch process handles the monthly calculation and distribution of interest for both Savings Accounts and Fixed Deposits based on a standard 30-day cycle.

- **Priority:** High.

4.4.2 Stimulus/Response Sequences

1. **Stimulus:** The System Administrator triggers the "Run Monthly Interest Engine" from the central simulator dashboard.
2. **Response:** The system iterates through all active Savings Accounts and FDs, calculating the interest accrued over the 30-day cycle.
3. **Response:** The system automatically credits the calculated interest to the respective Savings Accounts, logging each credit as a distinct transaction.

4.4.3 Functional Requirements

- **REQ-4.1:** The system shall process a 30-day period as the standard monthly cycle for all interest calculations.
- **REQ-4.2:** The system shall credit Fixed Deposit interest directly into the customer's linked Savings Account.
- **REQ-4.3:** The system shall log every automated interest credit as an individual, separate transaction in the activity log.
- **REQ-4.4:** The interest engine batch process must complete execution within a 15-minute maintenance window.
- **REQ-4.5:** The system shall ensure that the batch interest calculation is fully atomic.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** The web application shall load dashboard pages, generate transaction summaries, and complete standard queries (such as checking account balances or viewing active fixed deposits) within 2.0 seconds under normal load conditions (up to 50 concurrent regional service agents).
- **Transaction Processing:** Critical write operations, including deposits, withdrawals, and fixed deposit creations and it must be fully processed, validated, and committed to the backend database within 1.0 second to ensure smooth operations during busy business hours across branches in Sri Lanka.
- **Concurrency:** The central system infrastructure must comfortably support at least 50 concurrent users (comprising regional service agents and system administrators) accessing the system simultaneously without noticeable performance degradation.
- **Interest Calculation Batch Performance:** The central system's scheduled monthly interest distribution job (processing 30-day cycles for savings accounts and fixed deposits across all active plans) must complete execution for all accounts within a 15-minute maintenance window during off-peak hours.

5.2 Safety Requirements

- **Data Integrity and ACID Compliance:** Because the system handles monetary funds and interest accruals for B-Trust microfinance customers, all financial transactions (deposits, withdrawals, and monthly interest distributions) must be executed using database transactions wrapped with appropriate stored procedures and triggers to strictly guarantee ACID (Atomicity, Consistency, Isolation, Durability) properties, preventing partial updates or lost funds.
- **Overdraft Prevention:** The system must incorporate automated validation checks (via database constraints and procedural logic) to completely prevent account overdrafts, ensuring that withdrawals can only be performed if the account holder meets minimum balance and eligibility criteria.
- **Transaction Logging and Audit Trails:** Every financial operation must be automatically and immutably logged with precise timestamps, transaction types, and unique reference numbers to safeguard against financial discrepancies and provide a reliable audit trail for management reports

5.3 Security Requirements

- **Role-Based Access Control (RBAC):** Access to system features and data must be strictly restricted based on user roles (e.g., regional service agents, branch managers, and system administrators). Agents can only manage customers and accounts assigned to their respective branches.
- **User Authentication:** All users must securely authenticate with unique credentials before gaining access to the web application. Passwords must be securely hashed and salted using industry-standard algorithms (such as bcrypt).
- **Data Privacy and Protection:** Customer personal data, account balances, and transaction histories must be protected against unauthorized access or exposure, complying with standard data privacy practices relevant to financial institutions in Sri Lanka

5.4 Software Quality Attributes

- **Reliability:** The system shall maintain an uptime of 99.9% during standard banking business hours, ensuring continuous availability for regional service agents performing daily deposit and withdrawal operations.
- **Correctness:** All interest calculations (e.g., Children at 12%, Teen at 11%, Adult at 10%, Senior at 13%, Joint at 7%, and Fixed Deposits up to 15%) must execute with 100% mathematical precision according to defined monthly cycles, avoiding rounding or distribution errors.
- **Usability:** Given that regional service agents will use a lightweight UI for QA validation and daily operations, the interface must be intuitive, requiring minimal training, with clear error messages and responsive layouts.
- **Maintainability:** The backend database must utilize clean relational designs with clearly defined foreign keys, primary keys, and modular stored procedures, enabling future developers or DBAs to easily update business rules or generate management reports.

5.5 Business Rules

- **Branch and Agent Hierarchy:** B-Trust operates a series of service branches, with each branch managing a specific team of banking agents. Customers register at a branch and are assigned to a designated agent.
- **Account Eligibility and Minimum Balances:** Savings accounts have strict minimum balance entry rules: Children accounts require no minimum balance, Teen requires

LKR 500, Adult requires LKR 1000, Senior requires LKR 1000, and Joint accounts require LKR 5000. Withdrawals are prohibited if these criteria are unmet or if an overdraft would occur.

- **Fixed Deposit Constraints:** Customers may only open a Fixed Deposit if they hold an active Savings Account, and each customer is strictly limited to one FD per Savings Account. FD options include 6 months (13%), 1 year (14%), and 3 years (15%), with interest calculated monthly over a 30-day cycle and credited directly to the linked savings account as a separate transaction.
- **Mandatory Management Reports:** The system must successfully generate five key management reports upon request: agent-wise transaction totals/values, account-wise transaction summaries and current balances, active FDs with next payout dates, monthly interest distribution summaries by account type, and comprehensive customer activity reports.

6. Other Requirements

- **Database Requirements:** The system backend must be implemented using a robust relational database management system, utilizing proper indexing on frequently queried columns (such as account IDs, transaction dates, and agent IDs) to optimize report generation performance. Sample data must be pre-loaded into the database, including at least 5 agents, 3 branches, 15 customers (with at least 2 joint accounts), 10 fixed deposits, and 100 mixed transactions.
- **Internationalization and Localization:** Currency fields must natively support Sri Lankan Rupees (LKR), and date-time formats must align with local regional standards.

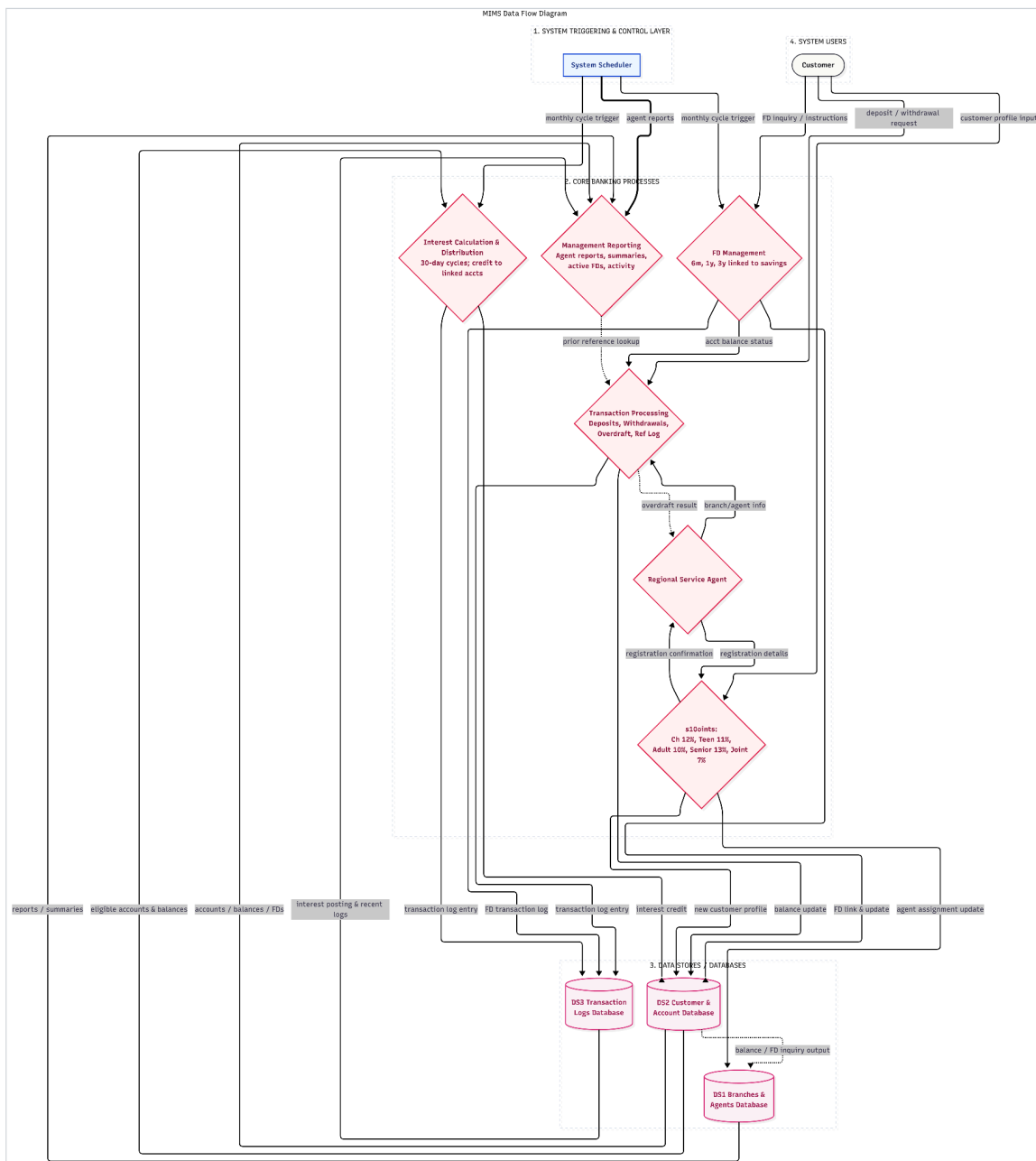
Appendix A: Glossary

- **ACID:** Atomicity, Consistency, Isolation, Durability — a set of properties that guarantee database transactions are processed reliably.
- **B-Trust:** A small private microfinance bank operating across several districts in Sri Lanka.
- **FD:** Fixed Deposit — a financial deposit account that pays a fixed interest rate over a specified term (6 months, 1 year, or 3 years).
- **MIMS:** Microbanking and Interest Management System.
- **SRS:** Software Requirements Specification.
- **UI:** User Interface.

Appendix B: Analysis Models

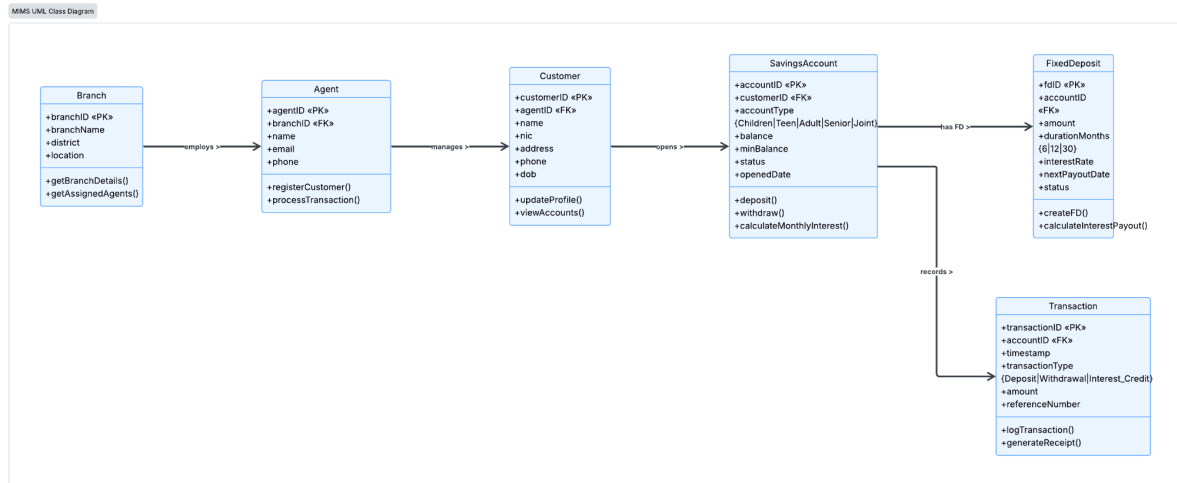
- **data flow diagram**

Below is the structured Data Flow Diagram (DFD) representation for the **Microbanking and Interest Management System (MIMS)**, broken down from a high-level context view to a detailed process level.



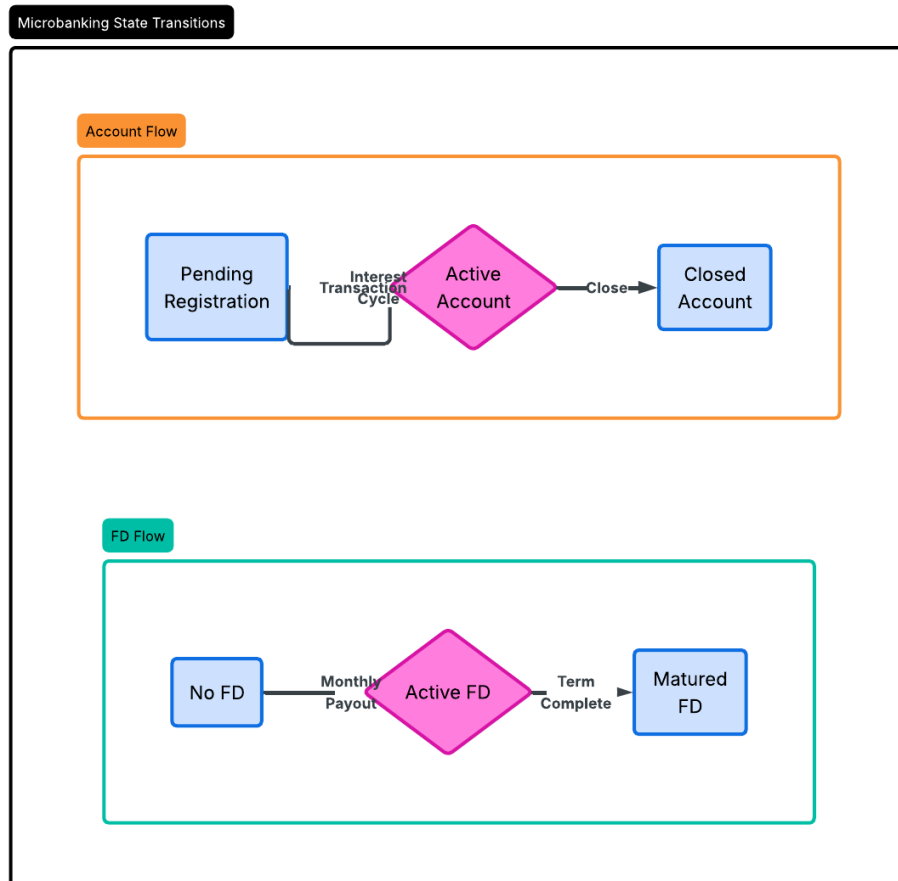
- **class diagram**

Below is the structured technical representation for the **Microbanking and Interest Management System (MIMS)** class diagram, broken down from an architectural class overview to detailed entity relationships, attributes, and methods.



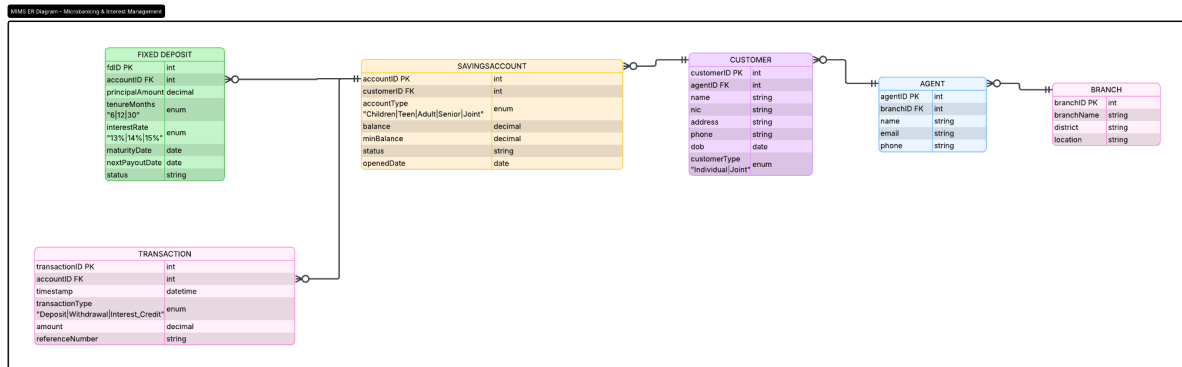
- **state-transition diagrams**

Below is the structured technical representation for the **Microbanking and Interest Management System (MIMS)** state-transition model, detailing the lifecycle states, trigger events, and transitions for both Savings Accounts and Fixed Deposits within the web application.



- **entity-relationship diagrams**

Below is the structured technical representation for the Microbanking and Interest Management System (MIMS) Entity-Relationship (ER) model, detailing the core database entities, structural attributes, referential constraints, and business rules enforced across the web application.



Appendix C: To Be Determined List

- **TBD-1:** Final hosting server specifications and cloud deployment environment details.
- **TBD-2:** Exact visual styling guidelines and CSS framework preferences for the lightweight UI web application.